Financial World-Model Runtime: Compiling Asynchronous Market Evidence for LLM Agents

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Abstract

Large language models and autonomous agents increasingly *consume* market evidence, yet raw financial feeds are asynchronous, intermittently missing, and quality-heterogeneous. Generative world models assume a usable observation stream; financial agents often lack that stream. We propose a *financial world-model runtime*: a point-in-time (PIT) compiler that maps raw multi-band evidence into an AI-consumable world state with explicit quality scores and abstention. The primitive is an epistemic observation $O_{j,t} = (x, \tau, q, g, r)$; a compilation operator Π_t yields a usable filtration $\mathcal{F}_t^{\mathrm{AI}}$; world-quality indices (WMI/ACWMI) gate actions. On a cryptocurrency PIT panel, vintaged macro and stablecoin-flow *content* changes a transparent consumer rule relative to same-input momentum (ΔCE 0.334, p=0.034), with leave-one-band-out (LOBO) losses dominated by content. A Compiled-versus-Raw protocol tests the agent interface. Trading metrics validate that the compiled world has content; they do not claim a strategy Holy Grail.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

world models, LLM agents, information-set compilation, selective prediction, point-in-time, cryptocurrency, abstention

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1 Introduction

AI agents that analyze or trade markets do not observe a clean panel. Venue fragmentation, derivatives surfaces, on-chain flows, news, and macro vintages arrive on incompatible clocks [11, 12]. Literatures on machine-learning asset pricing [5, 9, 13] and on generative world models [6, 7, 10] largely assume that observations are already aligned for learning or control. For LLM and tool-using agents [1, 16], the prior systems problem is different: *compile* asynchronous evidence into a world state that can be analyzed, refused when thin, and acted on when ready.

Thesis. We build a financial *world-model runtime*: an *observation-side* world model that compiles raw market evidence into a quality-tagged state for AI consumers. Trading statistics are instruments that show this world has nonempty content—not a claim of unconditional strategy alpha, and not a Dreamer-style learned dynamics model [7].

Contributions.

(1) **Semantics.** Epistemic observations, compilation operator Π_t, a delay/noise/missingness reconstruction bound, WMI/ACWMI, world-conditional abstention, and a LOBO content/gating decomposition.
(2) **Runtime.** A reproducible PIT multi-band system exposing quality-tagged bundles, availability shocks O_t, and a Compiled-versus-Raw consumer protocol (frozen prompts, temperature 0, abstain allowed).
(3) **Validation.** On a real PIT archive, compiled macro/alternative content yields a significant OOS certainty-equivalent (CE) gap versus same-input momentum, with content-dominated LOBO ablations and a long-span audit ruling out hidden return-rule alpha.

2 Related Work

World models in AI.. World models learn compact states and often dynamics for imagination and control [6, 7, 10]. We address the complementary problem in markets: constructing a PIT-safe, quality-tagged state *before* any dynamics model or LLM policy is applied.

LLM agents in finance. LLM workflows call tools and act on streams [1, 16]. Without a compiled world interface, agents face stale, ungated, or role-incoherent evidence. Our runtime supplies that interface; we evaluate a protocol, not live vendor alpha.

Selective prediction. Refusing to predict can be optimal under noise [3, 4]. We tie abstention to world quality rather than classifier confidence alone.

PIT measurement and crypto structure. Macro vintages [2] and crypto fragmentation [11, 12] motivate multi-band compilation. Inference uses bootstrap and snooping discipline [8, 14, 15].

3 Financial World-Model Runtime

Definition 1 (Epistemic observation). *An AI-consumable observation is*

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Identical x with different (τ, q, g, r) are different world objects.

Definition 2 (Compilation operator). *Let $\mathcal{F}_t^{\mathrm{raw}} = \sigma(\{X_{j,\tau}^{\mathrm{obs}}\})$. The compiler is $\Pi_t = B_t \circ M_t \circ A_t$ (align, missingness/honesty gates, band assembly), the world state is $W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$, and $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$.*

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World quality. Let $a_{k,t} \in \{0,1\}$ indicate whether band k meets an AI-usable threshold among K critical bands, $d_{j,t}$ a freshness penalty, and $m_{j,t}$ an incorrect main-view injection. Define

$$B_t = \frac{1}{K}\sum_{k=1}^{K} a_{k,t}, \qquad U_t = \exp\Big(-\sum_j \omega_j d_{j,t}\Big),$$
$$H_t = 1 - \frac{1}{J}\sum_j m_{j,t}, \qquad \mathrm{WMI}_t = B_t\, U_t\, H_t. \tag{1}$$

A geometric ACWMI aggregates readiness factors with cascade/regime context for selective gating. Production thresholds are frozen on an in-sample grid and never retuned out of sample.

Reconstruction bound. Latent states evolve as $S_{t+1} = F(S_t, \eta_{t+1})$; source j observes a lagged map $X^{\mathrm{obs}}_{j,t} = h_j(S_{t-\ell_{j,t}}) + \nu_{j,t}$. Under Lipschitz h_j and bounded increments, reconstruction error admits

$$\|\tilde{S}_t - S_t\| \le C_1 \sum_j \omega_j \ell_{j,t} + C_2 \sum_j \omega_j \|\nu_{j,t}\| + C_3 \sum_j \omega_j (1 - z_{j,t}), \tag{2}$$

with availability $z_{j,t}$. Latest-snapshot, TTL, and readiness gates shrink the three terms.

Proposition 1 (Compilation ≠ feature expansion). *Enlarging $\mathcal{F}^{\mathrm{raw}}$ without a well-defined Π_t need not enlarge decision-relevant $\mathcal{F}^{\mathrm{AI}}$: stale or role-incoherent evidence can raise raw span while lowering usable world quality via H_t and U_t.*

Proposition 2 (World-conditional abstention). *With action set including* abstain, *if every non-abstain action has expected loss above a world-dependent cost $c_{\mathrm{abs}}(W_t)$, the Bayes action is abstain. With costs weakly decreasing in* WMI, *abstention is a lower set in world quality [3].*

Definition 3 (LOBO content vs. gating). *Let $V(I)$ be OOS CE of a fixed action rule given information I. Band k's value $\widehat{\mathrm{MIG}}_k = V(I) - V(I \setminus E_k)$ decomposes into a* content *channel (changes the action map) and a* gating *channel (changes readiness/abstention only). Empirically we report a telescoping identity under the headline ungated rule.*

Remark 1 (Scope vs. generative WMs). *We do* not *claim a learned $p(s_{t+1} \mid s_t, a_t)$. This paper delivers a* state compiler + abstention runtime, *complementary to generative world models that assume a clean observation stream [6, 7].*

4 System

Figure 1 summarizes the anonymized prototype. Collectors ingest exchange, macro, alternative, and (when available) news/on-chain/options/tokenomics series into vintage-aware stores. A Band-PIT service reconstructs band readiness under a *previous-close* decision clock: positions earning calendar-day return r_t use only information known at $(t-1)$ 23:59. Content features are likewise vintaged with `available_at` $\le t$ and then lagged one day. The runtime computes readiness, WMI/ACWMI, availability shocks O_t, and quality-tagged AI bundles consumed by rule-based or LLM agents.

Bundle schema (AI-consumable state). Table 1 lists the fields exposed to consumers. Evidence-action rate (EAR) checks that every non-abstain action binds to named evidence. Bundles disclose whether paper engines or production proxies supplied (S, C).

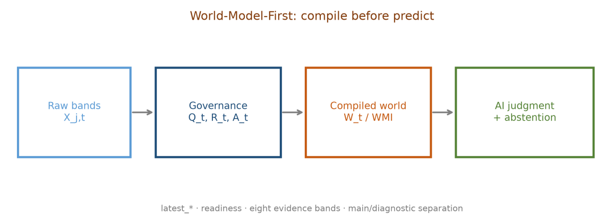

Figure 1: Observation-side financial world-model runtime: raw multi-band evidence is compiled into a quality-tagged world bundle for LLM / rule consumers with abstention.

Table 1: World-bundle schema served to AI consumers.

Field group	Contents
Identity	decision clock, asset universe
Bands	status, age, ready flags per band
Content	macro_tilt, alt_tilt, vintages
Quality	B, U, H, WMI, ACWMI
Engines	(S, C) + input-source disclosure
Shocks	availability O_t, outage markers
Audit	evidence IDs, EAR bindings

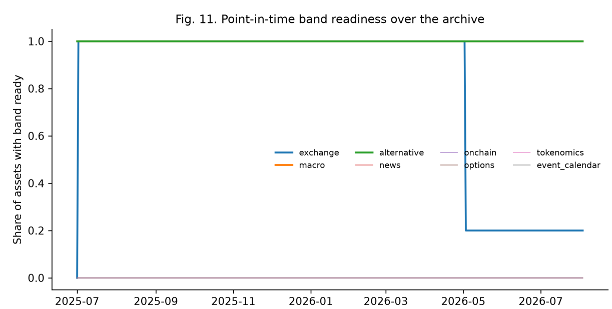

Figure 2: Point-in-time band readiness over the archive (durable bands dominate).

Runtime loop (conceptual). At each decision clock: pull latest PIT-safe observations $\{O_{j,t}\}$ → update band readiness and ages → compute $(B, U, H, \mathrm{WMI}, \mathrm{ACWMI})$ and content tilts → emit world bundle W_t → consumer returns $a_t \in \{\pm 1, 0, \text{abstain}\}$ → log evidence bindings. This loop is the system contribution: a silent feature store without quality, abstention, and Compiled/Raw contrast is not a world-model runtime.

Compiled-versus-Raw protocol. For consumer m,

$$\Delta_m = V_m(\text{Compiled}) - V_m(\text{Raw}),$$

with frozen prompts, temperature 0, and action schema {bullish, bearish, neutral, ab
Offline mocks keep the protocol executable without API keys; live vendors may replace mocks without changing the estimand. No live LLM trading alpha is claimed.

Table 2: OOS economic value on the PIT panel (content validation).

Policy	Sharpe	CE	Abstain	N
Always long	−1.399	−1.157	0.00	200
Momentum always	0.101	−0.202	0.00	200
Thick ungated (mech.)	0.767	0.132	0.075	200
ACWMI (IS-frozen)	0.654	0.076	0.127	200

Durable historical bands in the present archive are {exchange, macro, alte with empirical ready rates approximately 0.80/1.00/1.00; news/on-chain/options/tokenomics remain right-censored and are not identifying.

5 Experiments

5.1 Setup

PIT panel: ~399 days, 10 liquid crypto names, previous-close clock, IS/OOS 200/200. Metrics: annualized Sharpe and CRRA CE (γ=2). Inference: circular block bootstrap (n=999, block length 5) with stationary-bootstrap confirmation [14].

Transparent content rule. Band content enters the action map directly (not only as a gate). PIT-safe tilts use vintaged `available_at`:

$$\text{macro\_tilt}_t = \begin{cases} +1 & \Delta_5\text{VIX} < 0 \wedge \Delta_5\text{DXY} < 0 \text{ (risk-on)}, \\ -1 & \Delta_5\text{VIX} > 0 \wedge \Delta_5\text{DXY} > 0 \text{ (risk-off)}, \\ 0 & \text{otherwise or band not ready}, \end{cases}$$

$$\text{alt\_tilt}_t = \text{sign}(\text{stablecoin 7d net supply change}),\ 0 \text{ if not ready.}$$

The deterministic rule shares all return-based inputs with momentum and adds content:

$$\text{(R1) crisis} \wedge \text{cascade\_p} \geq 0.60 \Rightarrow -1,$$
$$\text{(R2) trend} \wedge \text{mom5} > 0 \wedge \text{cascade\_p} < 0.45 \wedge \text{macro\_tilt} \geq 0 \Rightarrow +1,$$
$$\text{(R2b) range} \wedge \text{macro\_tilt} > 0 \wedge \text{alt\_tilt} > 0 \wedge \text{mom5} \geq 0 \Rightarrow +1,$$
$$\text{(R3) else sign(mom5), long vetoed when both tilts risk-off.}$$

A zero signal is an intentional stand-aside. Because tilts are forced to zero when their band is not PIT-ready, LOBO deletes *content*, not only gating. The single pre-specified OOS contrast is **mechanism − momentum**.

5.2 RQ1: Does compiled content matter?

Table 2 reports the OOS horse-race. Always-long fails in this window (CE −1.157); momentum is nearly flat (Sharpe 0.101, CE −0.202); the content-aware rule earns Sharpe 0.767 / CE 0.132, standing aside on 7.5% of asset-days via content-driven vetoes. IS-frozen ACWMI is secondary (Sharpe 0.654, CE 0.076). Headline ΔCE=0.334 (p=0.034; stationary p=0.022; 95% CI [0.03, 0.68] excludes zero). At 10bps, ΔCE=0.333 (p=0.034) even though mechanism CE turns slightly negative (−0.019); at 25bps, ΔCE=0.332 (p=0.038). Absolute profitability is therefore fragile; relative content value is not (Figure 4). The gap is stable across $\gamma \in \{1, 2, 4, 6\}$ and a leave-one-asset jackknife.

Table 3: LOBO under ungated mechanism (OOS). Content share = 1.

Band dropped	ΔCE	p	Content share
Exchange	0.000	1.000	—
Macro	−0.334	0.034	1.0
Alternative	−0.334	0.034	1.0
Macro+alternative	−0.334	0.034	1.0

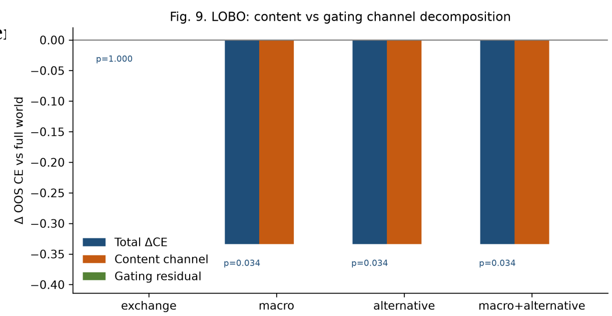

Figure 3: LOBO content vs. gating decomposition (ungated mechanism).

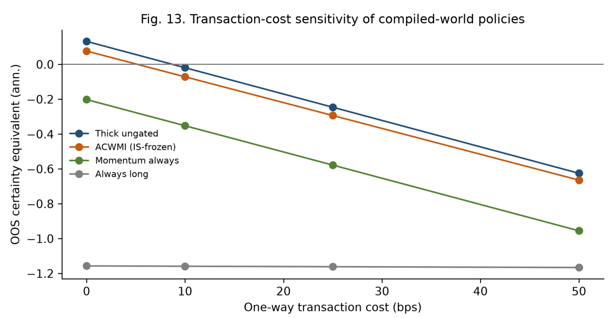

Figure 4: Cost sensitivity: absolute CE is fragile; relative content gap persists.

5.3 RQ2: Content vs. gating (LOBO)

Under the ungated mechanism, deleting macro or alternative collapses the rule onto momentum (ΔCE= − 0.334, p=0.034); content share of LOBO loss is 1.0 and gating residual is 0 (Table 3, Figure 3). Exchange status deletion costs 0 under the ungated rule. Band information changes *what* is traded, not only *when*. Nontrivial gating identification requires denser archives where abstention binds.

5.4 RQ3: Agent interface (Compiled vs. Raw)

Table 4 reports the offline mock suite. A compiled-aware mock abstains under thin compiled worlds (ΔCE= + 0.202 vs. raw); a momentum mock is invariant; a noisy mock can degrade. These are protocol smoke tests with evidence-binding EAR≈ 1, not unconditional LLM alpha.

Table 4: Compiled–Raw consumer protocol (offline mocks, OOS).

Consumer	ΔCE	Abs. (C)	Abs. (R)
mock-compiled-aware	+0.202	1.00	0.00
mock-momentum	0.000	0.00	0.00
mock-noisy	−0.036	0.00	0.00

5.5 RQ4: External audit

On 2017–2026 without vintaged band archives, the return-based core shows no hidden edge over momentum. Injecting joint *unsigned* content proxies (Yahoo VIX/DXY + DefiLlama stablecoin supply change) can change 157 of 6630 signals, but does *not* recreate the PIT CE gap versus no-content (ΔCE≈ -0.022, p=0.40). Macro-only proxies change zero signals, confirming two-band conjunction. Gains in Tables 2–3 load on compiled PIT content, not a secret return predictor.

5.6 Discussion

Read the tables as *content validation for the world-model runtime*, not as a mandate to trade frictionlessly. Enlarging usable I_t by vintaged macro/alternative content changes actions and raises OOS CE relative to same-input momentum (Props. 1, Def. 3). LOBO's content channel dominates gating under the ungated rule—partly by construction when content deletion collapses to momentum; that is exactly the content channel the runtime must expose. The agent protocol (Table 4) shows that Compiled versus Raw can change abstention behavior even for offline consumers. Menu-wide superiority versus always-long is *not* established (reality-check p=0.144); we do not claim it.

6 Limitations

News/on-chain/options/tokenomics lack durable history; full multi-band identification is future work. Absolute CE is cost-fragile. Consumer tests are protocol-first (mocks); live multi-vendor LLM evaluations remain open. WMI thresholds must be support-matched on sparse archives—production WMI<0.2 abstains essentially always here, consistent with Prop. 2 but not a completed selective-prediction horse-race. We provide an observation-side world-model runtime, not a full generative dynamics model $p(s_{t+1} \mid s_t, a_t)$.

7 Conclusion

Financial AI agents need a world to read. We formalize and implement a world-model runtime that compiles asynchronous evidence into an analyzable, abstention-aware, tradeable state $\mathcal{F}_t^{\mathrm{AI}} = \sigma(\Pi_t(\mathcal{F}_t^{\mathrm{raw}}))$, and show that compiled band content is economically nonempty under a disciplined PIT design. Trading numbers prove content; they do not proclaim a strategy Holy Grail. Generative dynamics and live multi-vendor LLM consumers are natural next layers on the same interface.

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